# DecoLive Jacket with Battery-free Dynamic Graphics

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**Figure 1: The DecoLive Jacket. (a) The jacket with the closed flower visualisation activated; (b) Activation gestures ; (c) Two states of the patterns on the DecoLive Jacket. Top: Closed flower, Bottom: Open flower.**

## ABSTRACT

We present the design and prototype of *the DecoLive Jacket*, a dynamic clothing that allows its wearer to switch between opened and closed states of flower patterns as subtle indication of availability for social interactions. The Jacket embodies two *electrochromic (EC) displays* and *NFC-based wireless energy harvesting components*, enabling battery-free operation. The wearer controls the garment's patterning by placing a smartphone in one of the two pockets of the jacket, activating either the open or closed flower visualization, and expressing a subtle message about their willingness to interact with others. Our implementation demonstrates the potential for battery-free EC displays on aesthetic wearables to enable dynamic control of the non-verbal messages communicated by clothing.

## CCS CONCEPTS

• **Human-centered computing → Ubiquitous and mobile computing**; **Interaction paradigms**; **Interaction design**; *Displays and imagers*.

## KEYWORDS

wearable computing, aesthetic computing, wearable displays, ambient displays, social wearables, electrochromic displays, design

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## 1 INTRODUCTION

The choices of color, style and materials of the garments send symbolic messages to others, and, consequentially, influences the impressions and behaviors of others towards the wearer [6, 13]. In that sense, symbolically communicative attributes of clothing have taken the interest of smart clothing designers. Whilst much work has been presented demonstrating dynamic changes in the visual qualities of garments with, for instance using thermochromic materials [5, 15], LEDs [1, 2, 8], optical fibers [7] or electrohromic (EC) displays [10], the power requirements of such interactive technologies still poses a limitation for the general acceptability of wearables. Wearable designs with batteries embedded create the burdensome need of charging them. Also, as issues of energy efficiency and sustainability are increasingly important factors for the future ubiquitous computing design, exploration of zero energy approaches for wearable computing deserves further exploration.

Self-powered wearables have so far gained far less attention, yet some examples exist. Solar cell powered approaches have been suggested, e.g., [14], requiring large surfaces to achieve full energy autonomy. NFC powered displays was also explored, e.g., for powering a passive RFID system by bringing an NFC reader in proximity [12]. AlterWear [4] integrated small displays on wearables and used NFC energy harvesting without a battery or other storage. However, the displays used in AlterWear were off-the-shelf E-ink displays, which were of small size, non-flexible and rectangular, arguably, failing to blend in the aesthetic of the garment design.

Motivated by the battery challenges and the possibility of influencing social relations with the garments' appearance, we investigate battery-free dynamic graphics on clothes to subtly invite or defer social encounters. For this purpose, we designed and prototyped a dynamic clothing, the DecoLive Jacket, Figure 1. It integrates two decorative EC displays [9], custom-shaped as two states of a flower -open and closed- on both sides of the chest area. There are also two energy-harvesting NFC tags embedded in the jacket's pockets to enable battery-free control of two decorative screens. The wearer controls the symbolic message presented, simply by putting an NFC enabled smartphone in, or close to, either one of the pockets. Our work demonstrates the potential for battery-free EC displays on aesthetic wearables to enable dynamic control of the non-verbal messages communicated by clothing.

## 2 THE CONCEPT & IMPLEMENTATION

The DecoLive Jacket (Figure 1) is a dynamic clothing that allows the wearer to switch between states of its controllable patterns for indicating their social availability. It is designed as a unisex garment that is suitable for casual wear in public places.

For representing the willingness or the unwillingness of the wearer towards socializing, two states of the *Paisley-style flowers* blended with the overall aesthetic of the garment: A closed flower facing sideways to represent that the wearer is not willing to socialize and an open flower facing frontwards to convey the message that the wearer is available for social interaction. Two states Paisley flower symbolically are chosen as they are representative for social availability while the possibility of creating stems with embroidery details is appealing to blend the displays with the overall design. The patterns are achieved through the use of two custom-designed EC displays, which can switch between two visual states. A detailed production process of the EC displays can be found in [9]. Both flower displays include two states (Figure 1-c): *High-opacity flower state* and *low-opacity flower state* where only the stem graphic becomes visible. Depending on the polarity of the applied voltage, one screen can be set to the high opacity flower state while the other switches to the low opacity.

As the power source, our concept utilizes a phone integrated NFC reader. The energy was harvested from the phone, using two *NFC Tag 2 Click* boards placed behind the pockets. The boards supply power (2.9 V) to switch the EC displays when a smartphone is placed in the range of the tag. The wearer controls the states by putting their smartphone in one of the jackets' pockets or touching it to the embroidery details (Figure 1-b). These gestures enable the jacket-embedded NFC module to harvest energy from the smartphone and switch the display located above the pocket to the high opacity flower state, while the flower on the opposite side fades away (Figure 1-a). When, e.g., the wearer is in a café and wishes to uninterruptedly read a book, they may set the closed flower to visible. When they want to stop reading and be open to socializing, the open flower can then be activated. Due to the memory effect of EC displays, the state of the flower displays remains observable for approximately 20 minutes without the need for the phone.

The electrical connections (GND and 2.9V) between the NFC Tag 2 Click boards and displays were accomplished with thin, flexible wires (Figure 2). Two connections were also made between the screens with the same type of wires on the internal part of the jacket. This enables each NFC Tag to apply opposite polarities to each EC display, setting one flower pattern to high opacity and the other to a low opacity. Two diodes were used between to prevent current flow from one NFC tag to the other.

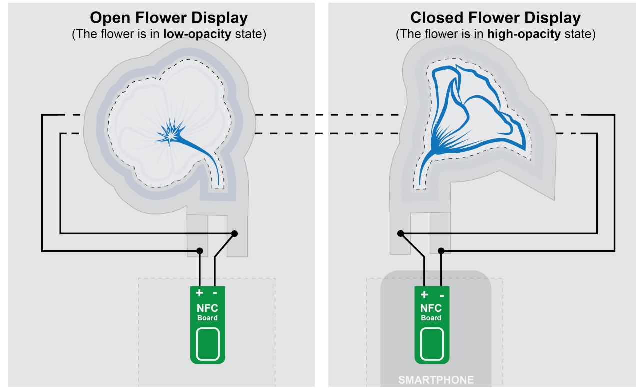


**Figure 2: Circuit diagram of the implementation.**

The prototype was made from a white polyester-cotton blend fabric, with blue 100% cotton in the collar and in the pockets. For embroidery, blue 100% merc. cotton yarn was used. The embroidery was designed to blend the technology into the jacket and to achieve a unique look. The embroidery on the pockets also indicates where the phone should be positioned to make the NFC connection to switch the state of the EC flower displays (Figure 1-b).

## 3 DISCUSSION AND CONCLUSION

When creating the DecoLive Jacket, we wanted to implement a battery-free technology solution which would let the users to control non-verbal messages on a jacket. Our target was to include unobtrusive and aesthetic display elements into the jacket. As pointed out by Devendorf et al., users do not want to feel that they are wearing a screen [3]. Here, non-illuminating EC display technology provides an appealing approach. Also, the free-form and flexible nature of these displays allows blending these screens with the other garment design elements, such as decorative stitching.

Furthermore, to enable battery-free interactivity, we used NFC harvesting technology, which has been utilized in wearable computer prototyping earlier both with garments and accessories [4], as well as with on-skin wearable technologies [11]. Using NFC technology as an energy source makes the interactive jacket easier to maintain as no batteries or chargers are required, which is an important factor from the practicality point of view with everyday wear. This battery-free solution is also more sustainable, as batteries are hazardous waste and contain toxic elements [12]. The energy-autonomous approach is also a conscious and important design choice towards more sustainable wearable computing.

All in all, coupling two technologies -EC displays, and NFC energy scavenging- together in a jacket design showcased the potential for battery-free displays on aesthetic wearables to enable dynamic control of the non-verbal communication with clothes.

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